Rebecca Tseng

SF Bay Area | San Diego, CA

PROFESSIONAL SUMMARY

Self-motivated undergraduate studying Bioinformatics with a strong foundation in both biology and computation. I am eager to apply my background in genomics, data analysis, and machine learning to contribute to biomedical innovation.

EDUCATION

University of California, San Diego — Anticipated: March 2027
Bachelor of Science, Bioinformatics
Minor: Computer Science
Related Coursework: Genomic Tools & Technologies, Statistics for Bioinformatics, Molecular Sequence Analysis, Genetics, Molecular Biology, Biological Databases, Bioinformatics Laboratory, Biochemistry

SKILLS & Certifications

Languages: Python, R, Java, Bash (Shell Scripting), Go, C, C++
Computational tools: GitHub, Vim, Jupyter Notebooks, AWS
Bioinformatics toolkit: Bioconductor, Seurat, Scanpy
Operating Systems: Windows, Linux, Mac
Human Subjects Protection (HSP) Group 1: Biomedical Research by CITI *(Expiration: 6/2029)*

EXPERIENCE

UC San Diego Health - Pediatrics — Jan 2024 - Present
Student Research Assistant - Kyle Gaulton's Lab

- Senior Thesis: Developed a gradient boosting model to distinguish molecular features of microglia in HIV
- Investigated pancreatic β-cell stressor responses in type 1 diabetes using multi-omics data
- Implemented differential expression, enrichment, and gene/pathway-level association analyses; visualized results to support biological insight

Memorial Sloan Kettering Cancer Center — June 2026 - Present
Computational Biology Intern - Ed Reznik's Lab

- Conducted genomic analysis of cachexia patients in a pancancer cohort
- Investigated circadian rhythm disruption from FitBit Actigraph data

Stanford University School of Medicine — June - August 2025
CREST Program Intern - Christina Curtis' Lab

- Performed single-cell spatial transcriptomics analyses on metastatic colorectal cancer samples from three patients
- Identified spatial organization and differences between primary and metastatic tumors using manual and automated cell typing and deep learning–based methods

PRESENTATIONS & ABSTRACTS

Machine Learning of Single Cell Genomics to Predict HIV-Exposed Microglia — June 2026
Saltman Quarterly, Vol. 3
Characterizing spatio-temporal differences in primary and metastatic colorectal cancer — August 2025
Canary CREST Symposium
Determining the role of pancreatic beta cell stress responses in type 1 diabetes — April 2025
UC San Diego Undergraduate Research Conference

AWARDS

Provost's Honors — Fall 2023, Winter 2024, Winter 2025, Fall 2025, Spring 2026